Syllabus

CCNA Topics

> ♨ **Strategy**
>
> Focus on **practical networking skills** that you'll use day-to-day in a networking internship/job. This covers helpdesk, NOC, junior network admin roles with a **cybersecurity career path** in mind.

📦 Module 1: Network Fundamentals ⭐ FOUNDATION – MUST MASTER

Critical Topics 🔥

- [ ] **OSI Model** – Know ALL 7 layers with examples
  - [ ] Physical – cables, signals
  - [ ] Data Link – MAC, switches, frames
  - [ ] Network – IP, routers, packets
  - [ ] Transport – TCP/UDP, segments
  - [ ] Session, Presentation, Application – protocols
  - [ ] **Be able to explain what happens at each layer**
- [ ] **TCP/IP Model** (4 layers)
- [ ] **TCP vs UDP** 🔥
  - [ ] Connection-oriented vs Connectionless
  - [ ] 3-way handshake (SYN, SYN-ACK, ACK)
  - [ ] Reliable vs Best-effort
  - [ ] Use cases for each
- [ ] **Common Protocols & Port Numbers** – Memorize these 🔥
  - [ ] HTTP (80), HTTPS (443)
  - [ ] DNS (53)

- [ ] DHCP (67/68)
- [ ] FTP (20/21), SFTP (22), FTPS (990)
- [ ] SSH (22), Telnet (23)
- [ ] SMTP (25), POP3 (110), IMAP (143)
- [ ] SNMP (161/162)
- [ ] RDP (3389)
- [ ] ICMP (ping, traceroute)
- [ ] Syslog (514)
- [ ] NTP (123)
- [ ] LDAP (389), LDAPS (636)

- [ ] **IPv4 Addressing & Subnetting** – CRITICAL SKILL 🔥🔥🔥
    - [ ] IP address structure (Network + Host)
    - [ ] Subnet masks & CIDR notation (/24, /25, etc.)
    - [ ] Calculate: Network ID, Broadcast IP, Usable host range
    - [ ] **Practice until you can do it in 30 seconds**
    - [ ] Private IP ranges (RFC 1918)
        - [ ] 10.0.0.0/8
        - [ ] 172.16.0.0/12
        - [ ] 192.168.0.0/16
    - [ ] Public vs Private IPs
    - [ ] APIPA (169.254.x.x) – what it means
    - [ ] Loopback (127.0.0.1)
- [ ] **IPv6 Basics**
    - [ ] Why IPv6 exists
    - [ ] Address format (hexadecimal, :: compression)
    - [ ] Link-local (fe80::), Global unicast, Unique local
    - [ ] EUI-64 addressing concept
- [ ] **MAC Addresses**
    - [ ] Format (48-bit, OUI + device ID)
    - [ ] How switches use MAC addresses
- [ ] **Unicast vs Broadcast vs Multicast**
- [ ] **Collision Domain vs Broadcast Domain**
- [ ] **Network Topologies** (Star, Mesh, Ring, Bus)

- [ ] **Ethernet Standards** (10/100/1000BASE-T, basic fiber)
- [ ] **Half-duplex vs Full-duplex**

Interview Questions You'll Get

- "Explain the OSI model"
- "What's the difference between TCP and UDP?"
- "Subnet 192.168.10.0/24 into 4 equal subnets"
- "What port does HTTPS use?"
- "What's a private IP address?"

📦 Module 2: Network Access (Switching & VLANs) ⭐ DAILY USE

Critical Topics 🔥

- [ ] **How Switches Work**
  - [ ] MAC address table (CAM table)
  - [ ] Learning, Flooding, Forwarding, Filtering
  - [ ] Frame switching methods
- [ ] **VLANs** – You'll configure these constantly 🔥🔥
  - [ ] What is a VLAN & why use them
  - [ ] VLAN ID (1-4094)
  - [ ] Default VLAN (VLAN 1)
  - [ ] Data VLAN vs Voice VLAN vs Management VLAN
  - [ ] Creating VLANs
  - [ ] Assigning ports to VLANs
- [ ] **Access Ports vs Trunk Ports** 🔥
  - [ ] Access = single VLAN
  - [ ] Trunk = multiple VLANs
  - [ ] 802.1Q tagging
  - [ ] Native VLAN concept
- [ ] **Inter-VLAN Routing**

- [ ] Router-on-a-stick (subinterfaces)
- [ ] Layer 3 switch (SVI)
- [ ] **Basic Switch Configuration** 🔥
  - [ ] Set hostname
  - [ ] Configure IP address (management VLAN)
  - [ ] Set enable secret password
  - [ ] Configure console password
  - [ ] Configure VTY lines (SSH/Telnet access)
  - [ ] Save configuration (copy run start)
  - [ ] Show commands (show run, show vlan, show mac address-table)
- [ ] **STP (Spanning Tree Protocol)** – Understand the purpose 🔥
  - [ ] Prevents Layer 2 loops
  - [ ] Root bridge election
  - [ ] Port states (Blocking, Listening, Learning, Forwarding)
  - [ ] RSTP (Rapid STP) – just know it's faster
  - [ ] PortFast – when to use it
  - [ ] BPDU Guard – basic concept
- [ ] **Port Security** – Basic implementation
  - [ ] MAC address limiting
  - [ ] Sticky MAC addresses
  - [ ] Violation modes (shutdown, restrict, protect)
- [ ] **Link Aggregation (EtherChannel)** – Basic concept
  - [ ] Combining multiple links
  - [ ] LACP vs PAgP (know the difference)
- [ ] **CDP & LLDP**
  - [ ] What they do (neighbor discovery)
  - [ ] Security concern (information disclosure)
  - [ ] How to disable

Practical Skills

- [ ] Console into a switch (using PuTTY/SecureCRT)
- [ ] Navigate Cisco IOS

- [ ] Create and configure VLANs
- [ ] Set up trunk links
- [ ] Verify connectivity (ping, show commands)
- [ ] Backup and restore configurations

Interview Questions You'll Get

- "What is a VLAN?"
- "Difference between access and trunk port?"
- "How does STP prevent loops?"
- "How would you check which devices are connected to a switch?"
- "Configure a VLAN on a switch" (hands-on)

📦 Module 3: IP Connectivity (Routing) ⭐ ESSENTIAL

Critical Topics 🔥

- [ ] **How Routers Work**
    - [ ] Route packets between networks
    - [ ] Routing table – how to read it
    - [ ] Longest prefix match
    - [ ] Administrative Distance (AD)
    - [ ] Metric
- [ ] **Default Gateway** – Critical concept 🔥
    - [ ] What it is
    - [ ] Why devices need it
    - [ ] How to configure it
- [ ] **Static Routing** 🔥
    - [ ] When to use it
    - [ ] How to configure
    - [ ] Next-hop vs exit interface
    - [ ] Default route (0.0.0.0/0)

- [ ] **Dynamic Routing Protocols** – Solid understanding needed 🔥
    - [ ] **Distance Vector vs Link State**
    - [ ] **Interior vs Exterior** (IGP vs EGP)
    - [ ] **RIP** (old, rarely used)
        - [ ] Hop count metric
        - [ ] Max 15 hops
        - [ ] RIPv1 vs RIPv2
    - [ ] **OSPF** (most common in enterprise) 🔥🔥
        - [ ] Link-state protocol
        - [ ] Areas (Area 0 = backbone)
        - [ ] Router ID
        - [ ] Hello packets
        - [ ] DR/BDR election (multi-access networks)
        - [ ] Metric = Cost (based on bandwidth)
        - [ ] Basic configuration
        - [ ] Wildcard masks in OSPF
    - [ ] **EIGRP** (Cisco proprietary)
        - [ ] Hybrid protocol
        - [ ] Uses bandwidth + delay
        - [ ] Fast convergence
        - [ ] Successor vs Feasible Successor
        - [ ] Basic configuration
    - [ ] **BGP** (Border Gateway Protocol) – Just the concept
        - [ ] Exterior Gateway Protocol
        - [ ] Used on the Internet
        - [ ] AS numbers
        - [ ] eBGP vs iBGP
- [ ] **Basic Router Configuration** 🔥
    - [ ] Set hostname
    - [ ] Configure interfaces (IP address, no shutdown)
    - [ ] Set enable secret
    - [ ] Configure console/VTY access
    - [ ] Configure routing (static or dynamic)

- [ ] Save configuration
    - [ ] Show commands (show ip route, show ip interface brief, show running-config)
- [ ] **Troubleshooting Tools** 🔥
    - [ ] **Ping** – tests connectivity
        - [ ] ICMP echo request/reply
        - [ ] What different responses mean
    - [ ] **Traceroute** – shows path
        - [ ] How it uses TTL
        - [ ] Reading traceroute output
    - [ ] **Show commands**
        - [ ] show ip route
        - [ ] show ip interface brief
        - [ ] show protocols
        - [ ] show running-config
- [ ] **Common Routing Issues**
    - [ ] Incorrect subnet mask
    - [ ] Wrong default gateway
    - [ ] Interface down
    - [ ] Missing routes
    - [ ] Routing loops

Practical Skills

- [ ] Read a routing table
- [ ] Configure static routes
- [ ] Configure OSPF on a router
- [ ] Troubleshoot connectivity issues using ping/traceroute
- [ ] Verify routing protocol operation

Interview Questions You'll Get

- "What's in a routing table?"
- "Explain how OSPF works"
- "Difference between static and dynamic routing?"

- "How would you troubleshoot why two networks can't communicate?"
- "Configure OSPF on this router" (hands-on)

📦 Module 4: IP Services ⭐ YOU'LL USE DAILY

Critical Topics 🔥

- [ ] **NAT/PAT** – Very important 🔥🔥
    - [ ] Why NAT exists (IPv4 address shortage)
    - [ ] **Static NAT** – One-to-one mapping
    - [ ] **Dynamic NAT** – Pool of public IPs
    - [ ] **PAT (NAT Overload)** – Many-to-one (most common)
        - [ ] How it uses port numbers
    - [ ] Inside Local, Inside Global, Outside Local, Outside Global
    - [ ] Basic NAT configuration
    - [ ] Verifying NAT (show ip nat translations)
- [ ] **DHCP** – You'll configure this often 🔥🔥
    - [ ] **DORA Process** (Discovery, Offer, Request, Acknowledgment)
    - [ ] DHCP Scope
    - [ ] Lease time
    - [ ] DHCP Relay (ip helper-address)
    - [ ] Configuring DHCP server on Cisco router
    - [ ] Configuring DHCP client on interface
    - [ ] Excluded addresses
    - [ ] Default gateway, DNS server options
    - [ ] Troubleshooting DHCP
- [ ] **DNS** – Critical service 🔥
    - [ ] How DNS resolution works
    - [ ] Recursive vs Iterative queries
    - [ ] DNS hierarchy (Root, TLD, Authoritative)
    - [ ] A record, AAAA, CNAME, MX, PTR, NS
    - [ ] Forward lookup vs Reverse lookup

- [ ] DNS caching
- [ ] Common DNS issues
- [ ] **NTP (Network Time Protocol)** 🔥
  - [ ] Why time synchronization matters
  - [ ] Stratum levels
  - [ ] Configuring NTP client
  - [ ] Verifying NTP (show ntp status)
  - [ ] Importance for logging & troubleshooting
- [ ] **Syslog** 🔥
  - [ ] Centralized logging
  - [ ] Severity levels (0-7)
    - [ ] 0 = Emergency
    - [ ] 1 = Alert
    - [ ] 2 = Critical
    - [ ] 3 = Error
    - [ ] 4 = Warning
    - [ ] 5 = Notification
    - [ ] 6 = Informational
    - [ ] 7 = Debugging
  - [ ] Configuring logging to syslog server
  - [ ] Local logging
  - [ ] Buffer, console, terminal logging
- [ ] **SNMP** 🔥
  - [ ] Network monitoring protocol
  - [ ] Manager vs Agent
  - [ ] MIB (Management Information Base)
  - [ ] OID (Object Identifier)
  - [ ] **SNMP versions**
    - [ ] v1 – no security
    - [ ] v2c – community strings (still weak)
    - [ ] v3 – authentication & encryption (use this!)
  - [ ] Read-only vs Read-write community
  - [ ] Basic SNMP configuration

- [ ] **SSH vs Telnet** 🔥
  - [ ] Telnet = cleartext (insecure)
  - [ ] SSH = encrypted (secure)
  - [ ] Configuring SSH on Cisco devices
  - [ ] SSH version 2
  - [ ] Generating crypto keys
- [ ] **FTP vs TFTP vs SFTP**
  - [ ] FTP – File Transfer Protocol (ports 20/21)
  - [ ] TFTP – Trivial FTP (port 69, no authentication)
  - [ ] SFTP – SSH FTP (port 22, secure)
  - [ ] When to use each
  - [ ] Backing up configurations via TFTP

Practical Skills

- [ ] Configure DHCP server on router
- [ ] Set up NAT/PAT
- [ ] Configure NTP client
- [ ] Set up syslog logging
- [ ] Enable SSH and disable Telnet
- [ ] Backup config to TFTP server

Interview Questions You'll Get

- "Explain how DHCP works"
- "What's the difference between NAT and PAT?"
- "Why is NTP important?"
- "What are syslog severity levels?"
- "Configure DHCP on this router" (hands-on)

📦 Module 5: Security Fundamentals ⭐ CRITICAL FOR YOUR CAREER PATH

Critical Topics 🔥

- [ ] **CIA Triad** 🔥
    - [ ] Confidentiality
    - [ ] Integrity
    - [ ] Availability
- [ ] **AAA Framework** 🔥
    - [ ] Authentication – Who are you?
    - [ ] Authorization – What can you do?
    - [ ] Accounting – What did you do?
    - [ ] **RADIUS** – UDP, encrypts password only
    - [ ] **TACACS+** – TCP, encrypts entire payload, Cisco preferred
    - [ ] When to use each
- [ ] **Access Control Lists (ACLs)** – You'll use these 🔥🔥🔥
    - [ ] **Standard ACL** (1-99, 1300-1999)
        - [ ] Filters based on source IP only
        - [ ] Place close to destination
    - [ ] **Extended ACL** (100-199, 2000-2699)
        - [ ] Filters on source, destination, protocol, port
        - [ ] Place close to source
    - [ ] **Named ACL**
    - [ ] **Wildcard masks** – Inverse of subnet mask
    - [ ] **Implicit deny** – at the end of every ACL
    - [ ] **Inbound vs Outbound** application
    - [ ] ACL configuration and verification
    - [ ] Common mistakes (order of statements, implicit deny)
- [ ] **Firewall Concepts** 🔥
    - [ ] **Stateless** – filters based on rules only
    - [ ] **Stateful** – tracks connection state
    - [ ] Packet filtering
    - [ ] Next-Generation Firewall (NGFW)
        - [ ] Deep packet inspection
        - [ ] Application awareness
        - [ ] IPS integration
    - [ ] **DMZ** (Demilitarized Zone)

        - [ ] Buffer zone between internal and external
        - [ ] Where to place public-facing servers
- [ ] **VPN Fundamentals** 🔥🔥
    - [ ] **Site-to-Site VPN** – connects two networks
    - [ ] **Remote Access VPN** – connects user to network
    - [ ] **IPSec**
        - [ ] AH (Authentication Header) – integrity
        - [ ] ESP (Encapsulating Security Payload) – encryption
        - [ ] Transport mode vs Tunnel mode
        - [ ] IKE Phase 1 & Phase 2 (basic understanding)
    - [ ] **SSL/TLS VPN**
        - [ ] Browser-based
        - [ ] Easier for remote users
    - [ ] **GRE Tunnels** (Generic Routing Encapsulation)
        - [ ] Encapsulates many protocols
        - [ ] Not encrypted by default
        - [ ] Often used with IPSec
- [ ] **Wireless Security** 🔥
    - [ ] **Encryption Standards**
        - [ ] WEP – broken, don't use
        - [ ] WPA – better, still vulnerable
        - [ ] WPA2 – current standard (AES)
        - [ ] WPA3 – newest, best security
    - [ ] **Authentication Methods**
        - [ ] Personal (PSK – Pre-Shared Key)
        - [ ] Enterprise (802.1X with RADIUS)
    - [ ] SSID (Service Set Identifier)
    - [ ] Hidden SSID (security through obscurity – not effective)
    - [ ] MAC filtering (can be bypassed)
- [ ] **Port Security** 🔥
    - [ ] Limit MAC addresses per port
    - [ ] Static, Dynamic, Sticky MAC learning
    - [ ] Violation modes:

- [ ] Shutdown – disables port
    - [ ] Restrict – drops frames, logs
    - [ ] Protect – drops frames, no log
  - [ ] Configuration and verification
- [ ] **Layer 2 Security Threats** 🔥
  - [ ] **MAC Flooding** – overwhelms switch MAC table
  - [ ] **ARP Spoofing/Poisoning** – impersonates another device
  - [ ] **VLAN Hopping**
    - [ ] Switch spoofing
    - [ ] Double tagging
  - [ ] **DHCP Attacks**
    - [ ] DHCP Starvation
    - [ ] Rogue DHCP server
  - [ ] **STP Attacks** – becoming root bridge
  - [ ] **CDP/LLDP Reconnaissance** – info gathering
- [ ] **Layer 2 Security Mitigations** 🔥
  - [ ] **Port Security** – limits MAC addresses
  - [ ] **DHCP Snooping** – validates DHCP messages
    - [ ] Trusted vs untrusted ports
    - [ ] DHCP binding table
  - [ ] **Dynamic ARP Inspection (DAI)** – validates ARP packets
  - [ ] **IP Source Guard** – prevents IP spoofing
  - [ ] **BPDU Guard** – disables port if BPDU received
  - [ ] **Root Guard** – prevents unauthorized root bridge
  - [ ] Disable CDP/LLDP on untrusted ports
  - [ ] Disable unused ports
  - [ ] Change native VLAN from VLAN 1
- [ ] **Device Hardening** 🔥
  - [ ] Use strong passwords
  - [ ] Enable secret (not enable password)
  - [ ] Service password-encryption
  - [ ] SSH instead of Telnet
  - [ ] Disable HTTP, use HTTPS

- [ ] AAA authentication
- [ ] Login banners (legal notice)
- [ ] Disable unused services
- [ ] Regular updates/patches
- [ ] Backup configurations
- [ ] **IDS vs IPS** 🔥
  - [ ] **IDS** (Intrusion Detection System)
    - [ ] Monitors and alerts
    - [ ] Passive
  - [ ] **IPS** (Intrusion Prevention System)
    - [ ] Monitors and blocks
    - [ ] Inline
    - [ ] Active prevention
  - [ ] Signature-based vs Anomaly-based
  - [ ] Placement in network
- [ ] **802.1X (Port-Based NAC)** 🔥
  - [ ] Supplicant (client)
  - [ ] Authenticator (switch)
  - [ ] Authentication server (RADIUS)
  - [ ] EAP (Extensible Authentication Protocol)
  - [ ] Use cases

Practical Skills

- [ ] Configure standard and extended ACLs
- [ ] Set up port security
- [ ] Enable DHCP snooping
- [ ] Configure SSH access
- [ ] Implement device hardening steps

Interview Questions You'll Get

- "What's the CIA triad?"
- "Difference between RADIUS and TACACS+?"

- "Create an ACL to block HTTP traffic from 192.168.1.0/24"
- "What's DHCP snooping?"
- "How would you secure a switch?"
- "Explain stateful vs stateless firewall"

📦 Module 6: Wireless Networking (Light Touch)

Must-Know Topics

- [ ] **Wireless Standards** (just know the basics)
  - [ ] 802.11a/b/g/n/ac/ax
  - [ ] Speeds and frequencies (2.4 GHz vs 5 GHz)
- [ ] **Wireless LAN Components**
  - [ ] Access Point (AP)
  - [ ] Wireless LAN Controller (WLC)
  - [ ] Autonomous AP vs Lightweight AP
- [ ] **SSID & BSS**
- [ ] **CAPWAP Protocol** – communication between AP and WLC
- [ ] **Wireless Security** – covered in Module 5
- [ ] **Basic Troubleshooting**
  - [ ] Signal strength
  - [ ] Channel interference
  - [ ] Authentication issues

Skip/Low Priority

- ~~Wireless site surveys~~
- ~~RF propagation details~~
- ~~Advanced WLC configuration~~

📦 Module 7: Automation & Programmability (Awareness Level)

Must-Know Topics (Concepts Only)

- [ ] **Why Automation**
    - [ ] Consistency
    - [ ] Speed
    - [ ] Reduce human error
- [ ] **SDN (Software-Defined Networking)**
    - [ ] Control Plane vs Data Plane vs Management Plane
    - [ ] Centralized controller
    - [ ] OpenFlow (just know the name)
- [ ] **Controller-Based Networking**
    - [ ] Cisco DNA Center
    - [ ] Southbound vs Northbound APIs
- [ ] **REST API Basics**
    - [ ] What an API is
    - [ ] CRUD operations (Create, Read, Update, Delete)
    - [ ] HTTP methods (GET, POST, PUT, DELETE)
    - [ ] JSON format (how to read it)
- [ ] **Configuration Management Tools** (just know names & purpose)
    - [ ] Ansible
    - [ ] Puppet
    - [ ] Chef
    - [ ] SaltStack
- [ ] **Python for Network Automation** (nice to have)
    - [ ] Netmiko, Paramiko libraries
    - [ ] Not required for entry level, but good to mention interest

Skip/Low Priority

- ~~Deep Python programming~~
- ~~Advanced Ansible playbooks~~
- ~~Yang models~~

🎯 Skills You'll Actually Use in Internship/Job

Daily Tasks

- [ ] Troubleshoot connectivity issues (ping, traceroute)
- [ ] Configure VLANs on switches
- [ ] Set up trunk links
- [ ] Create and modify ACLs
- [ ] Configure DHCP scopes
- [ ] Set up NAT/PAT
- [ ] Add/remove users from network devices
- [ ] Monitor network performance
- [ ] Update device configurations
- [ ] Document network changes
- [ ] Respond to network alerts
- [ ] Assist with firewall rule requests
- [ ] Backup configurations
- [ ] Cable management and physical connectivity

Tools You'll Use

- [ ] **PuTTY / SecureCRT** – terminal emulation
- [ ] **Packet Tracer / GNS3** – lab simulations
- [ ] **Wireshark** – packet capture/analysis
- [ ] **TFTP/FTP server** – backup configs
- [ ] **Syslog server** – centralized logging
- [ ] **SNMP monitoring tools** – PRTG, SolarWinds, etc.
- [ ] **Ticketing systems** – ServiceNow, Jira, etc.
- [ ] **Documentation tools** – Confluence, SharePoint
- [ ] **Network diagrams** – Visio, Draw.io

Commands You'll Type 1000 Times

```
# Daily Use Commands
show running-config
```

```
show ip interface brief
show vlan brief
show ip route
show mac address-table
show cdp neighbors
show version
ping [IP]
traceroute [IP]

# Configuration Mode
configure terminal
hostname [name]
interface [type] [number]
ip address [IP] [mask]
no shutdown
exit / end

# Save Configuration
copy running-config startup-config
write memory (or just: wr)
```

Hands-On Projects for Resume

1. Home Lab – Document your setup
2. Multi-VLAN network with inter-VLAN routing
3. Site-to-site VPN between two routers
4. ACL implementation for network segmentation
5. DHCP server with multiple scopes
6. Network monitoring with Syslog + SNMP

```
# Daily Use Commands
show running-config
show ip interface brief
show vlan brief
show ip route
show mac address-table
show cdp neighbors
show version
ping [IP]
```

```
traceroute [IP]

# Configuration Mode
configure terminal
hostname [name]
interface [type] [number]
ip address [IP] [mask]
no shutdown
exit / end

# Save Configuration
copy running-config startup-config
write memory (or just: wr)

# VLAN Commands
vlan [number]
name [vlan-name]
switchport mode access
switchport access vlan [number]
switchport mode trunk

# ACL Commands
access-list [number] permit/deny [source] [wildcard]
ip access-list extended [name]
ip access-group [ACL] in/out

# Security Commands
enable secret [password]
service password-encryption
line vty 0 4
login local
transport input ssh
```